

GABLS1: measured scheme-native WENO flux

GABLS1 scheme-native WENO transport | 22 September 2026. Matched 12.5 m, 32³, 400 m, WENO9, one face, 300 s filter, factor one, seed 123, nine-hour LES. The old admitted factor-one covariance run is the control.

At the first interior face, the measured 8–9 h WENO numerical correction to u-momentum flux is $-0.000228 \text{ m}^2/\text{s}^2$, versus covariance $-0.007749 \text{ m}^2/\text{s}^2$ (2.95%). The heat correction is $1.4\text{e-}10 \text{ K m/s}$, effectively zero versus covariance -0.001472 K m/s . The saved covariance plus correction equals reconstructed flux at both diagnostic faces for momentum and scalars. These measured corrections are far below the assumed extra flux behind factors two and three in this case.

Mean 8–9 h first-face momentum viscosity falls 1.3214 to 1.2958 m^2/s (-1.9%); heat diffusivity rises 1.2160 to 1.2642 m^2/s (+4.0%). Integrated resolved TKE rises 3.8%, peak w^2 rises 3.1%, diagnosed boundary-layer height falls 7.4 m. One evolving realization cannot distinguish a systematic response from trajectory divergence. The profiles remain near the covariance control and show substantial coarse-run departures from the fixed 1 m LES median in resolved turbulence.

Source and data audit: immutable Breeze/Evaluation manifest SHA-256 72f1bc03 and 792 source-file hashes passed; paired initial-theta SHA-256 1f5db33f matched. H100 gate 7496 passed; science 7497 had durable exit zero and CASE_DONE at 32400 s. Raw output passed exact 1/19/541 initial/profile/series schedules, finite values, native 32/33-level heights, and flux decomposition on both faces. Figures use Julia. The reference is an LES median, not observations. Historical factor and control outputs were not rerun.

GABLS1: shear and surface-layer scaling

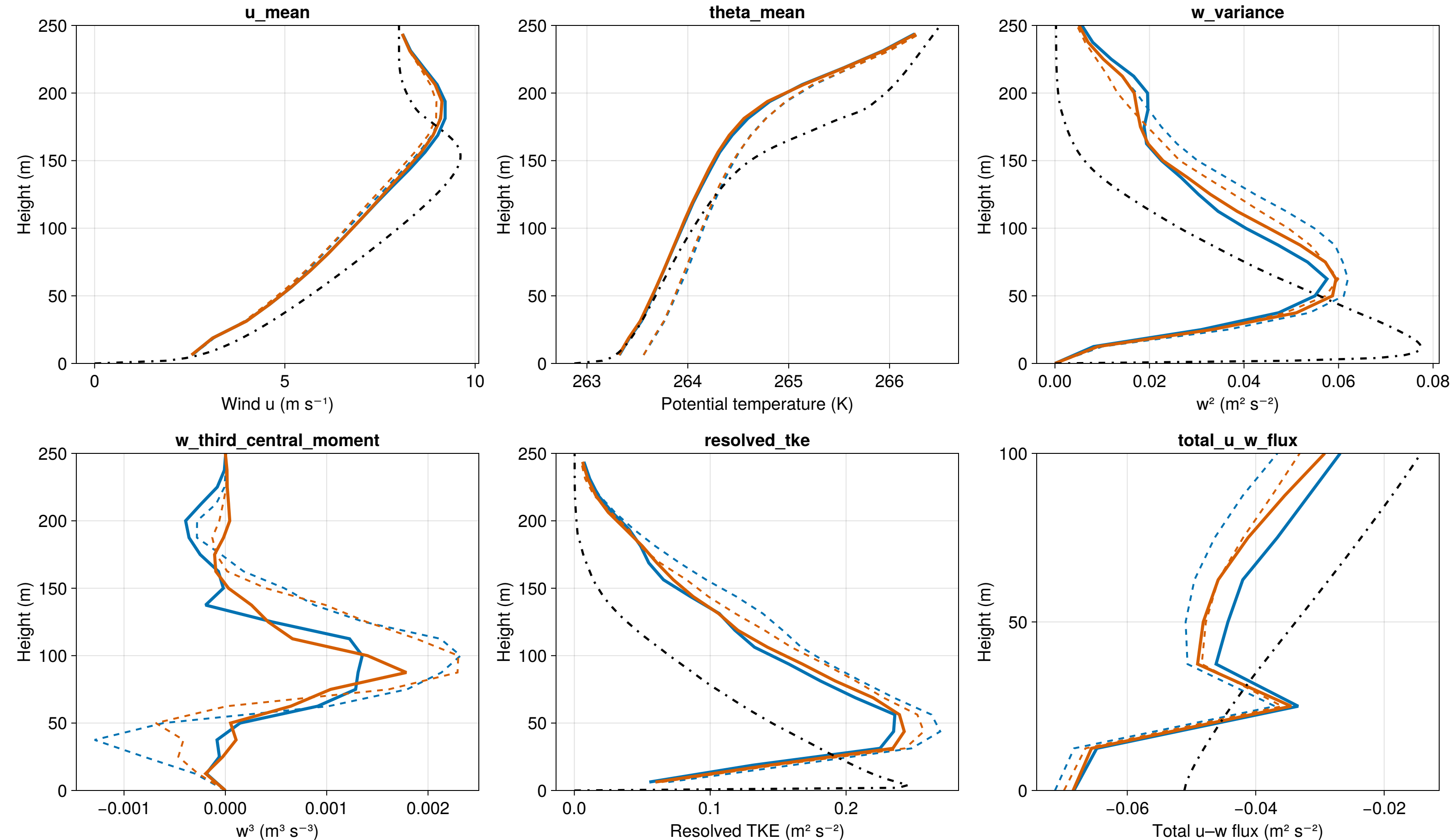
The final-hour mean horizontal-wind vector shear across 6.25-18.75 m is 0.0522 s^{-1} for scheme-native SLD and 0.0511 s^{-1} for covariance SLD, versus 0.1004 s^{-1} from the fixed 1 m median profiles; the no-closure case is high at 0.1861 s^{-1} . The pattern persists during 7-8 h, but differences change sign higher up. All model gradients use adjacent 12.5 m cell-center levels. Reference componentwise median u/v profiles are sampled at those heights and differentiated identically; this is not the median of member shears or a wall gradient.

A restricted comparison of the requested treatments gives first-layer vector shear 0.1861 s^{-1} without closure, 0.0522 s^{-1} with scheme-native SLD, and 0.1173 s^{-1} with Smagorinsky, versus 0.1004 s^{-1} from the fixed 1 m median profiles. Smagorinsky is from an earlier source revision and is contextual. The SLD deficit is concentrated near the surface; it is not a uniform shear bias over the full layer.

The 1 m median is not a neutral log-law profile over 2-30 m: horizontal speed has fitted log-height slope 1.123 m/s, whereas mean u^*/κ gives 0.654 m/s. Stable GABLS1 requires a positive Monin-Obukhov correction. That correction reduces, but does not eliminate, the departure for both the reference and scheme-native SLD. The wall coefficient couples the surface to the first velocity level; SLD acts at one interior face. Neither imposes a log or stable-similarity profile throughout the column. Time-averaged fluxes and independently median u/v components need not obey a pointwise similarity identity.

GABLS1: scheme-native transport versus covariance

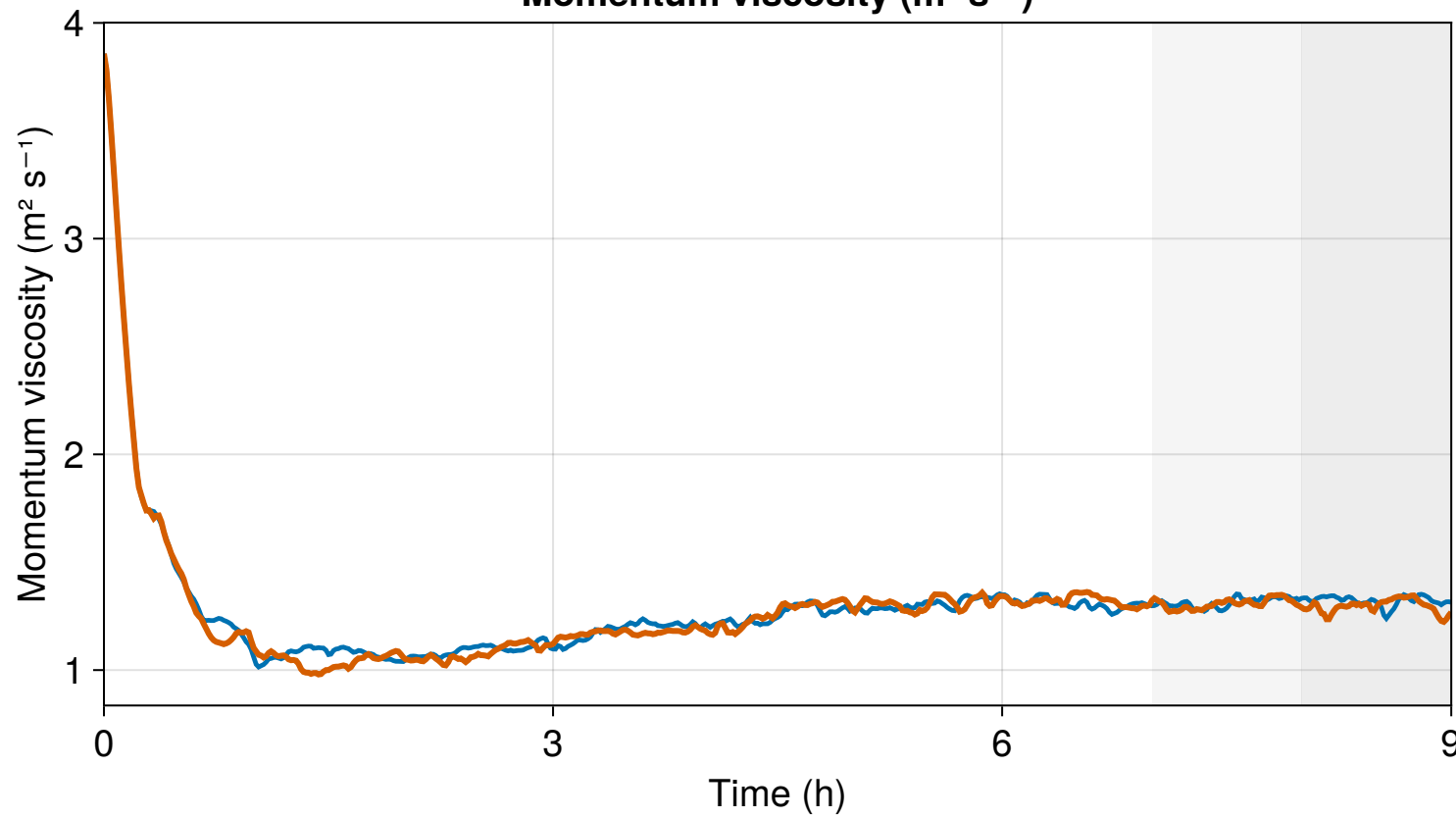
12.5 m · WENO9 · one face · 300 s · factor 1 · same seed; dashed 7–8 h, solid 8–9 h



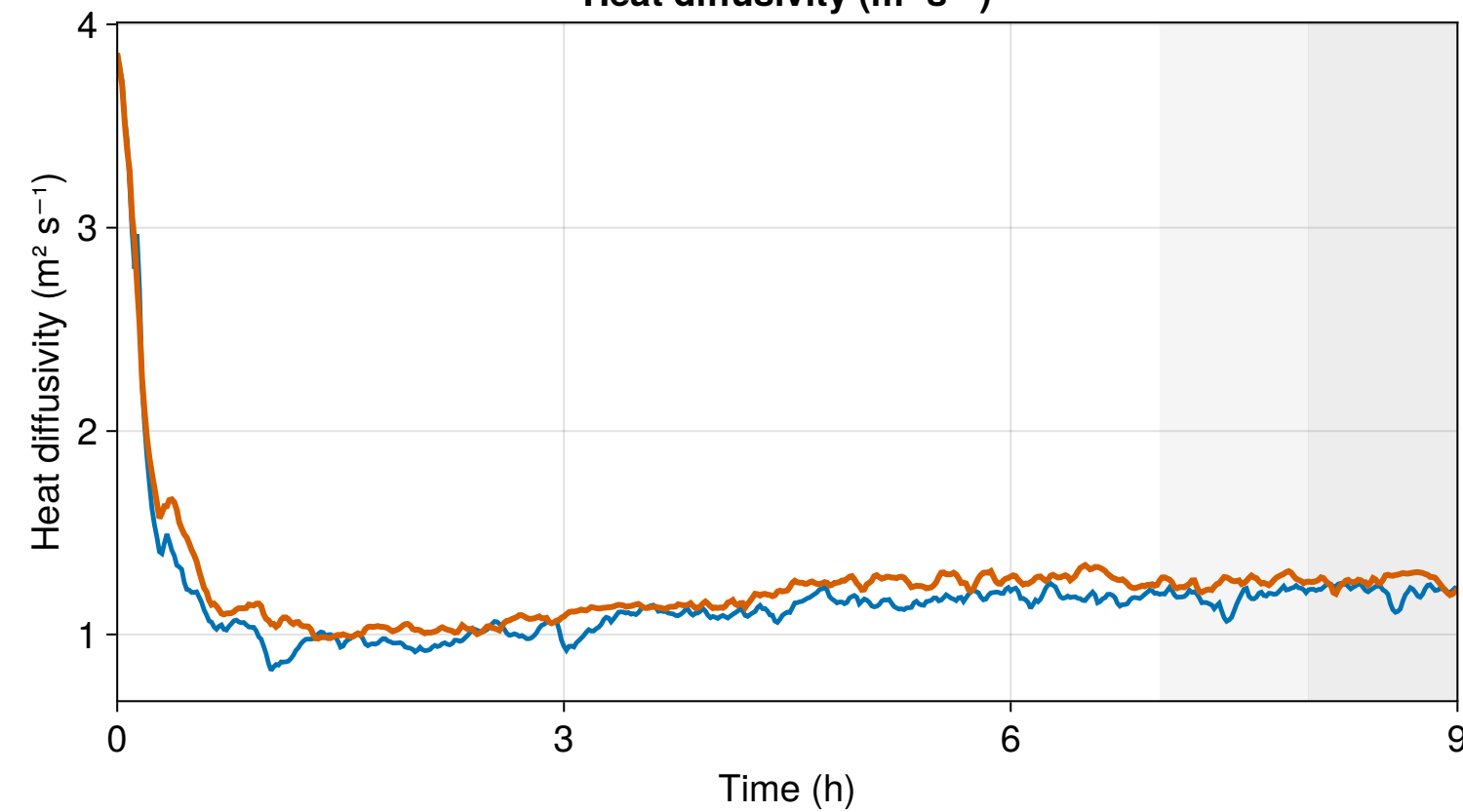
What the transport estimator changes at the first interior face

Same nine-hour forcing and discretization · 12.5 m supported face · shaded final two hours

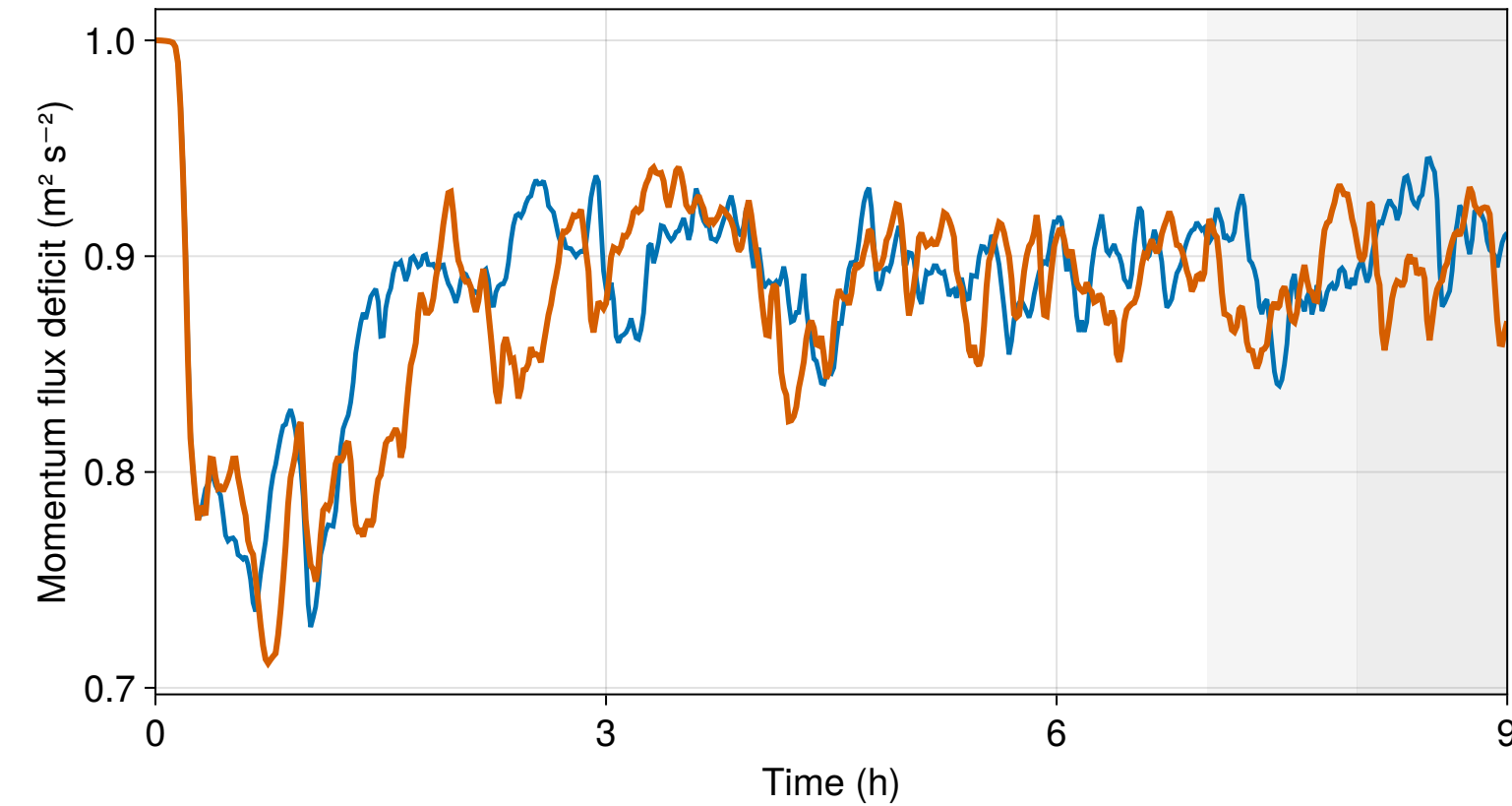
Momentum viscosity ($\text{m}^2 \text{s}^{-1}$)



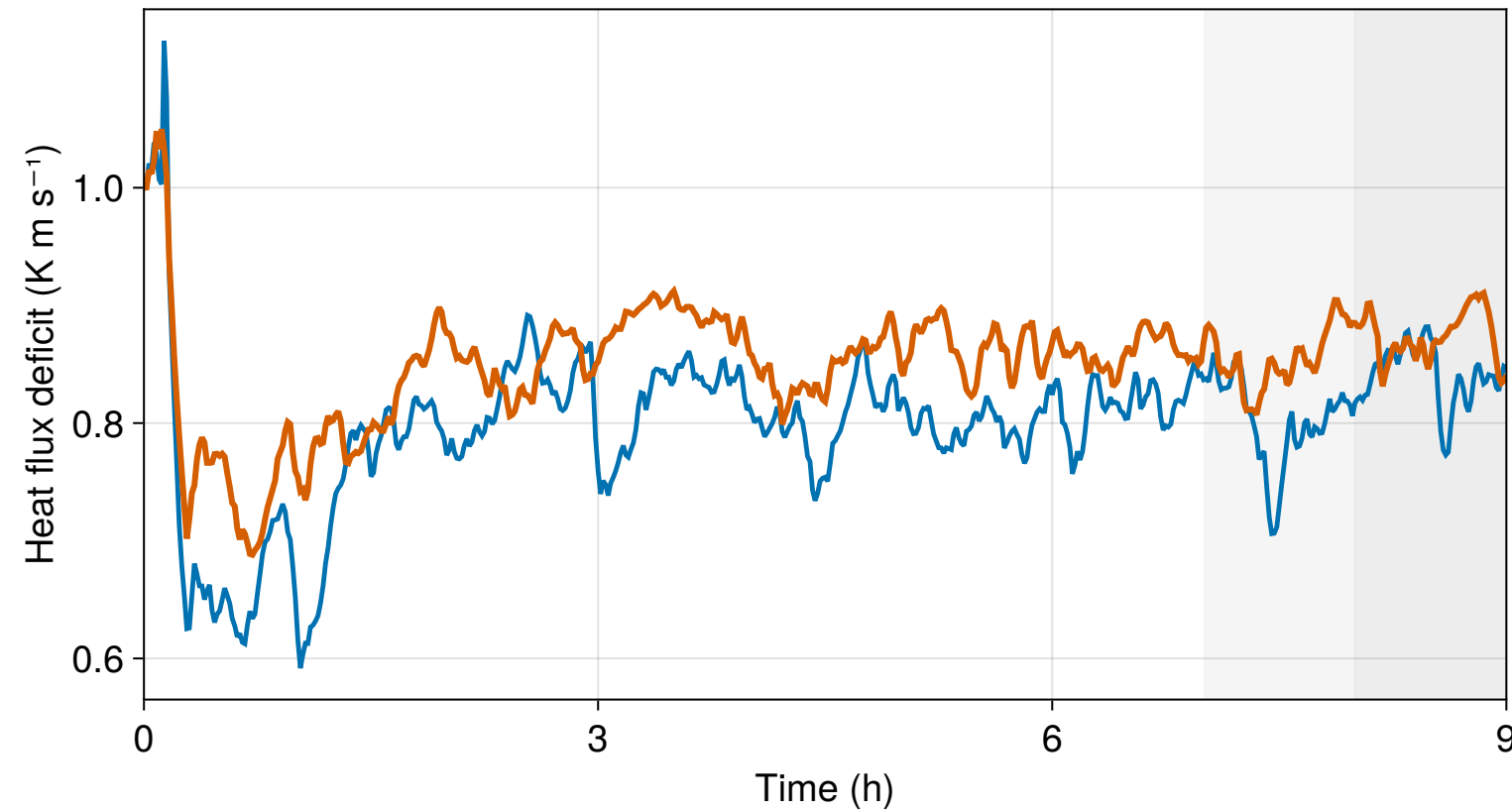
Heat diffusivity ($\text{m}^2 \text{s}^{-1}$)



Momentum flux deficit ($\text{m}^2 \text{s}^{-2}$)



Heat flux deficit (K m s^{-1})

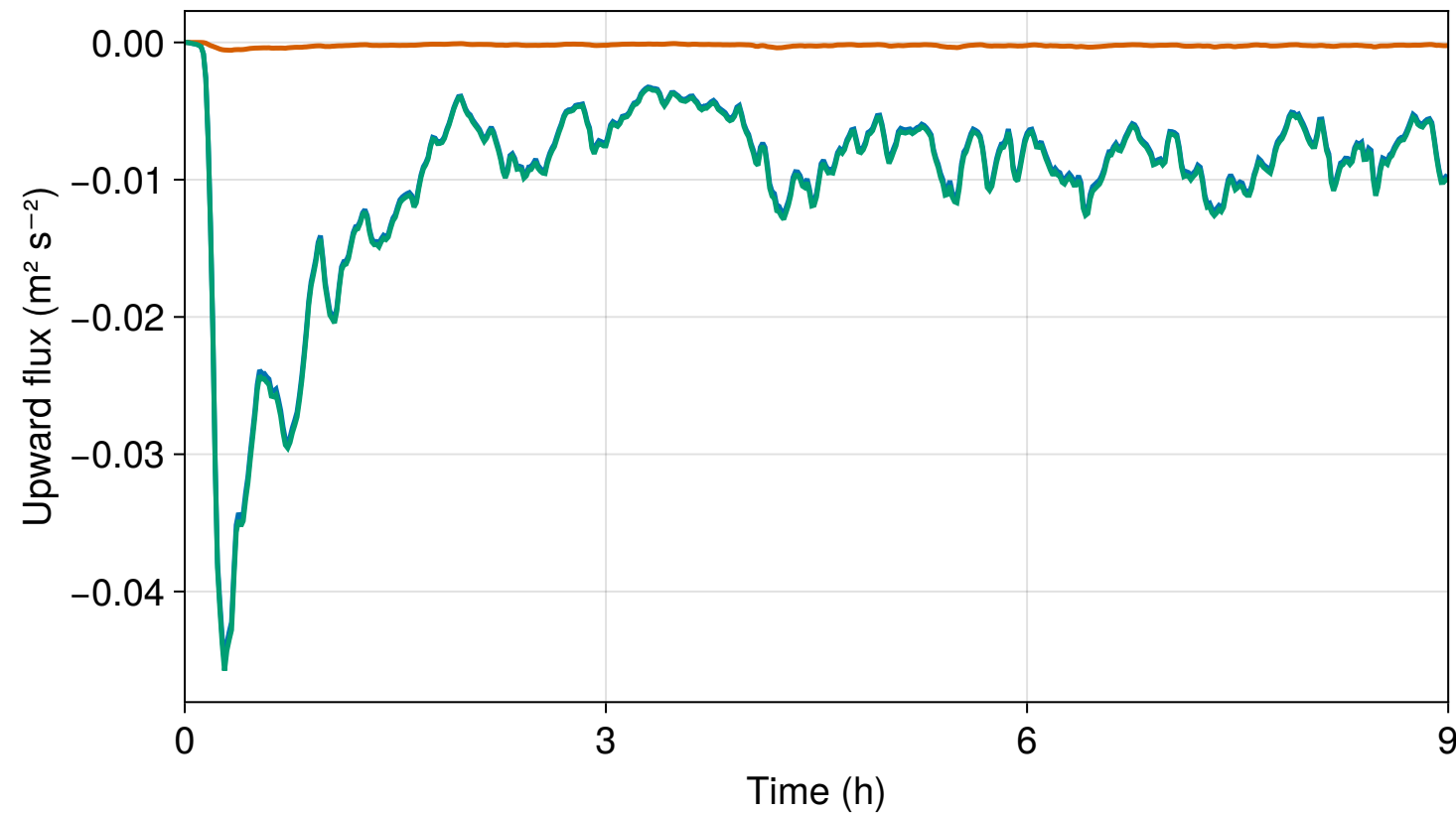


— Covariance control — Scheme-native

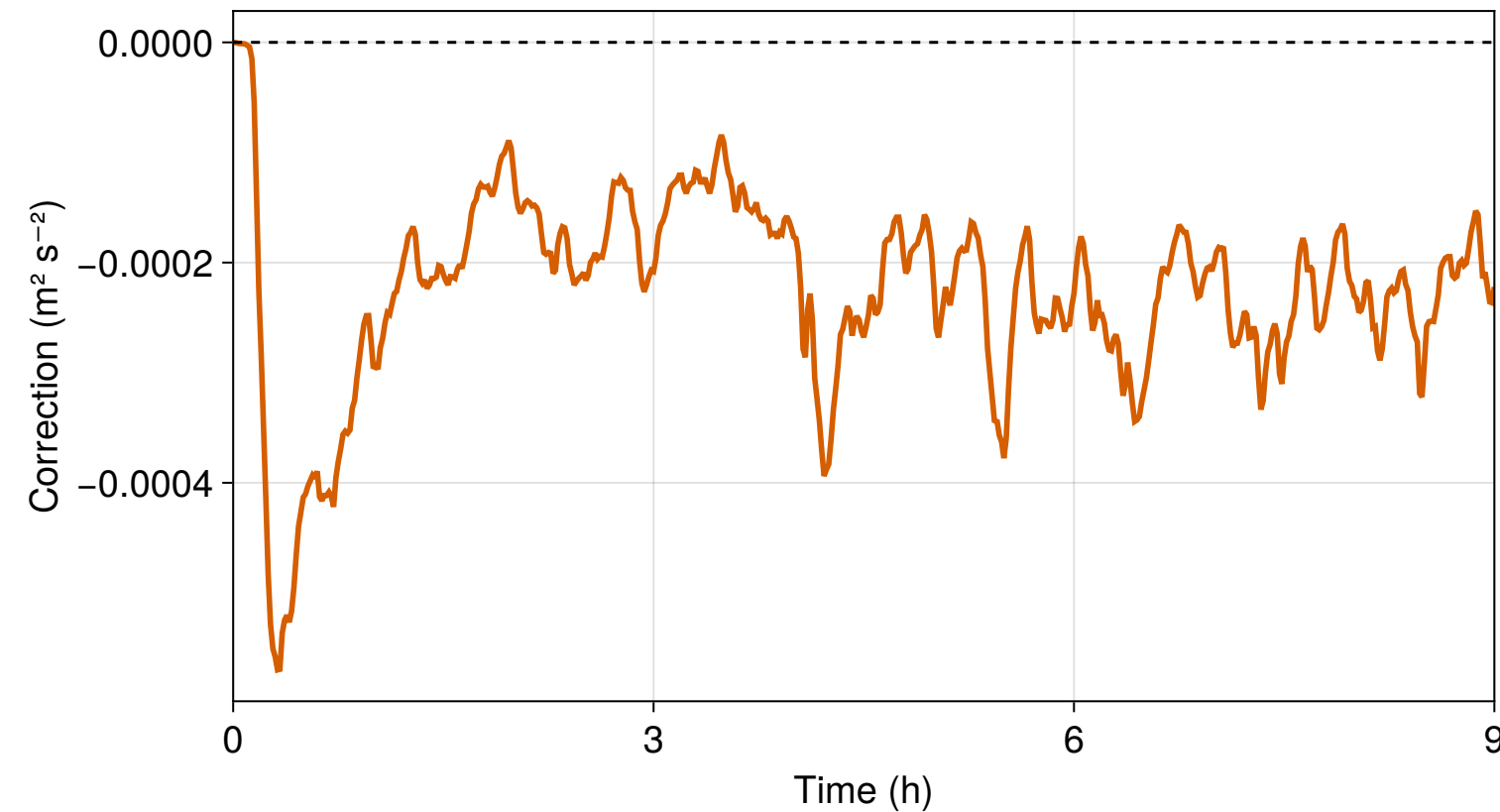
Measured WENO transport correction

Scheme-native run only · covariance + numerical correction = reconstructed flux

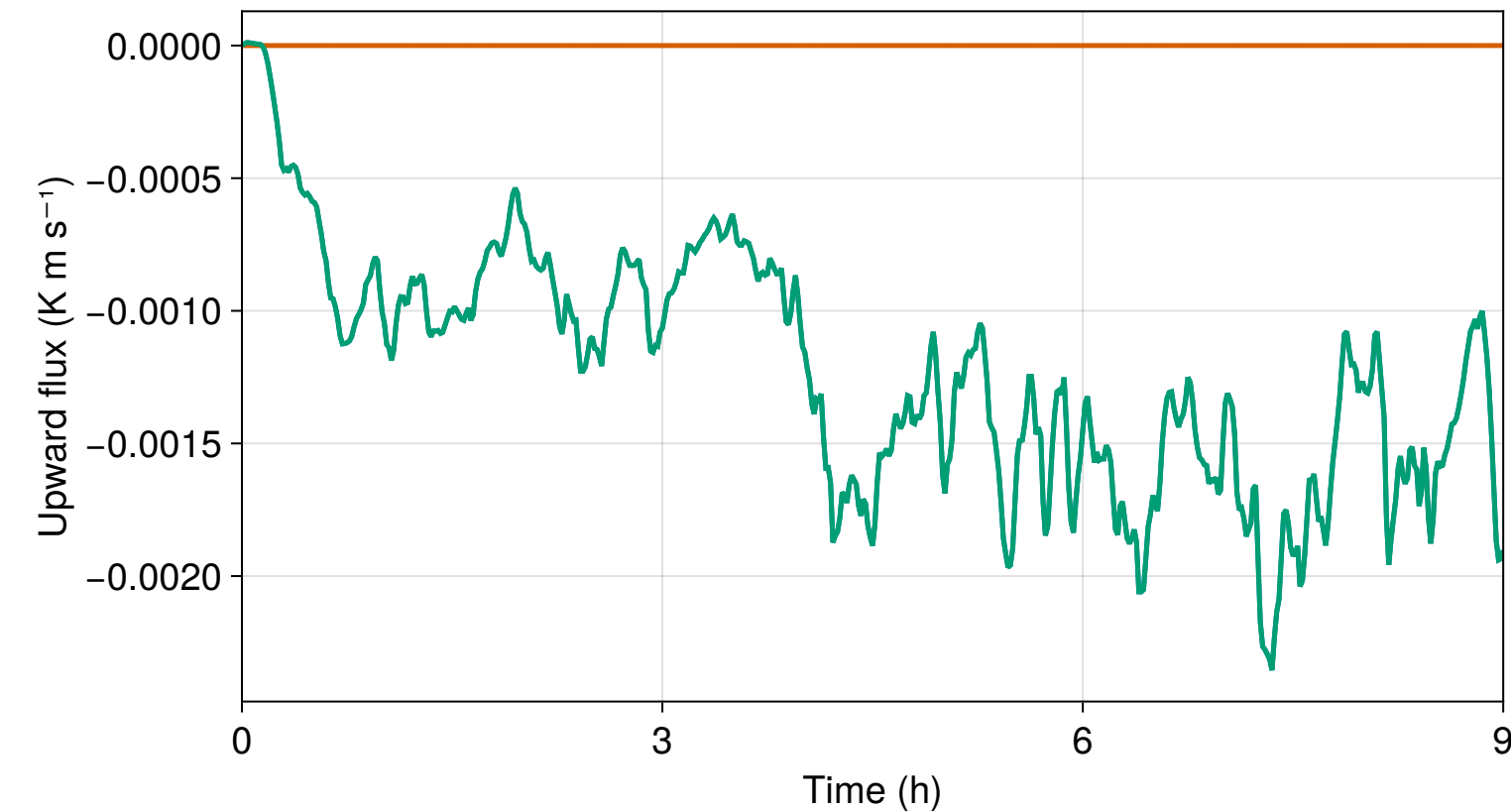
Momentum flux



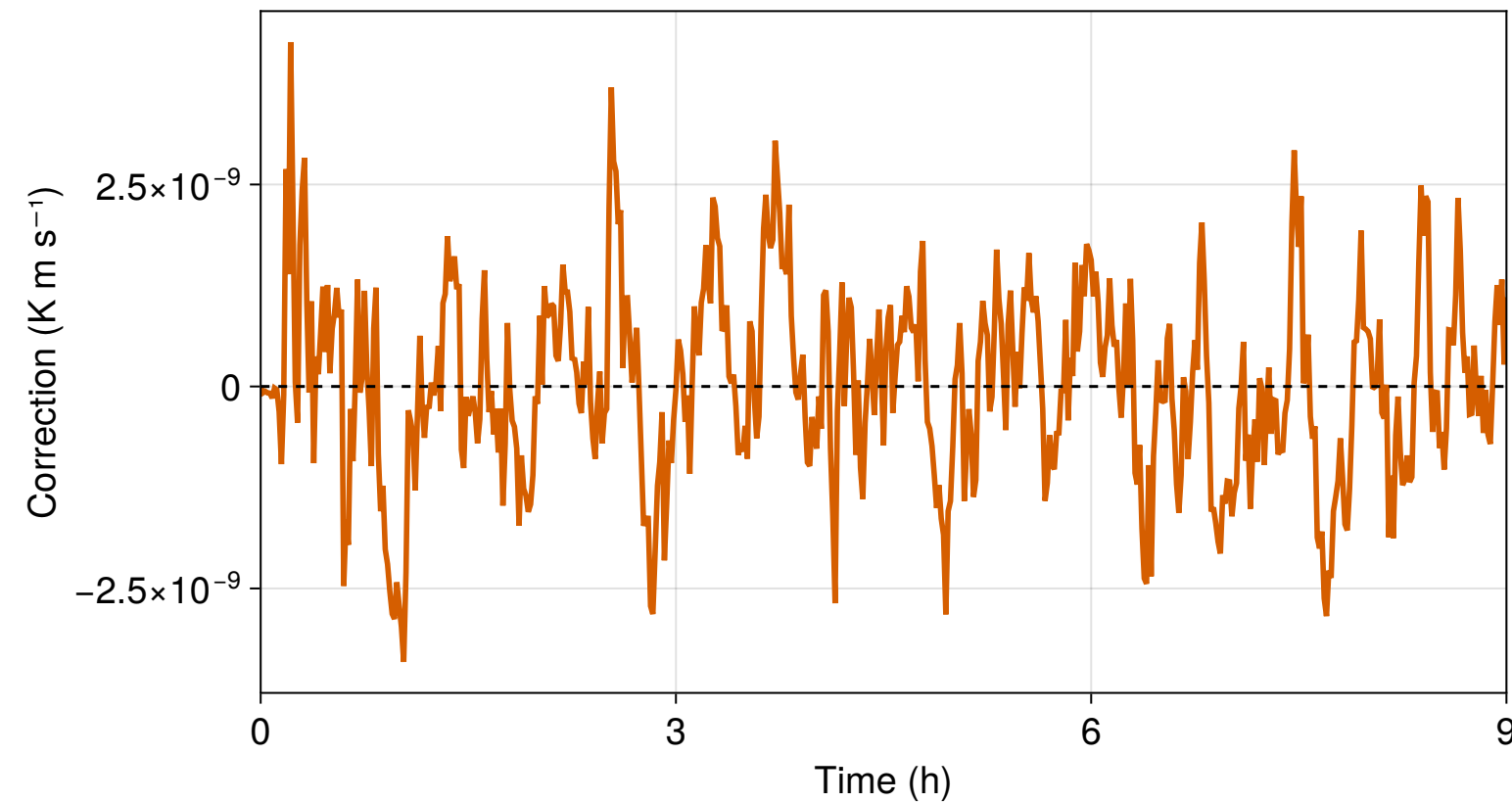
Momentum numerical correction



Heat flux



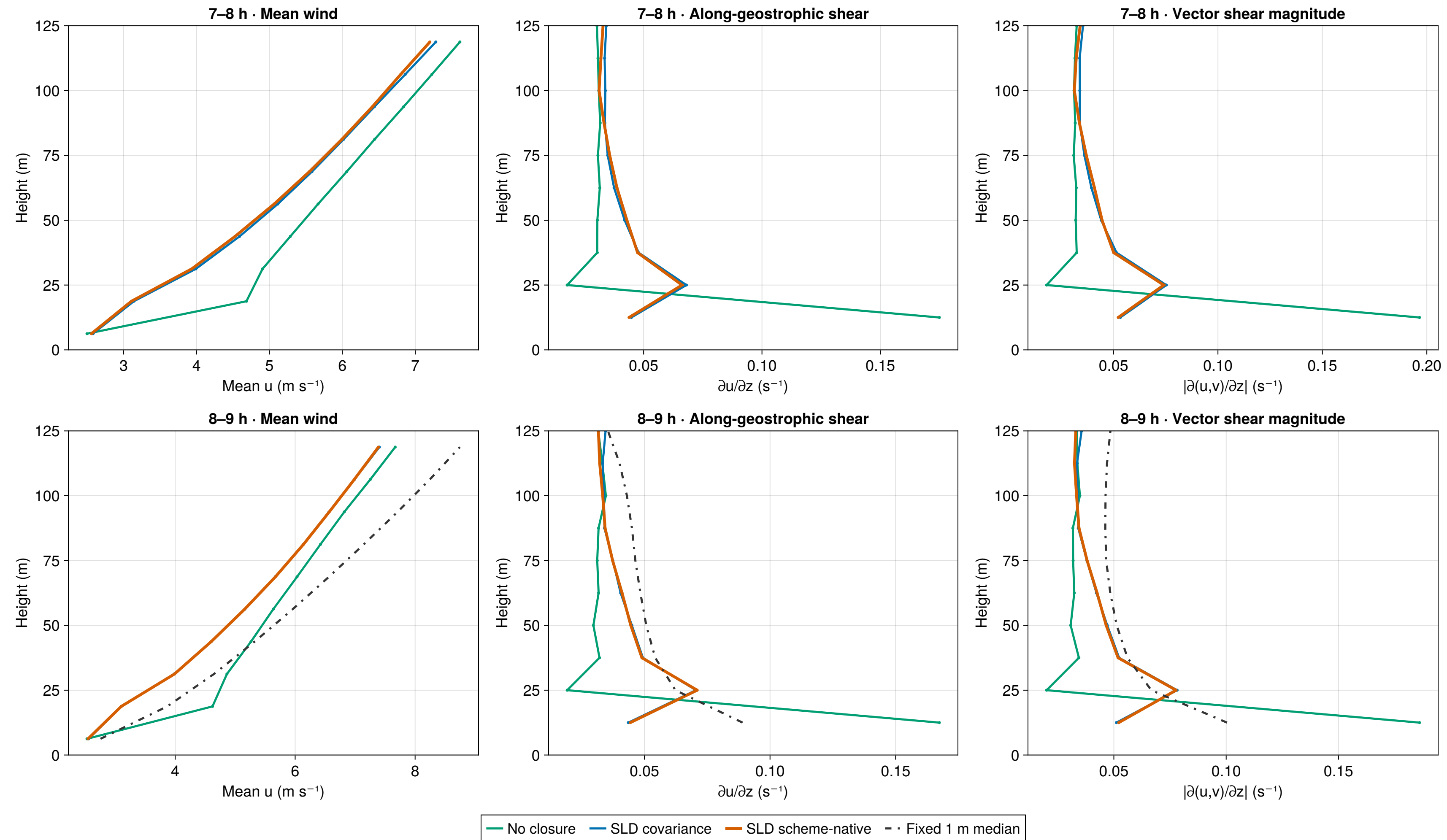
Heat numerical correction



— Covariance — Numerical correction — Reconstructed

GABLS1: vertical shear of the mean horizontal wind

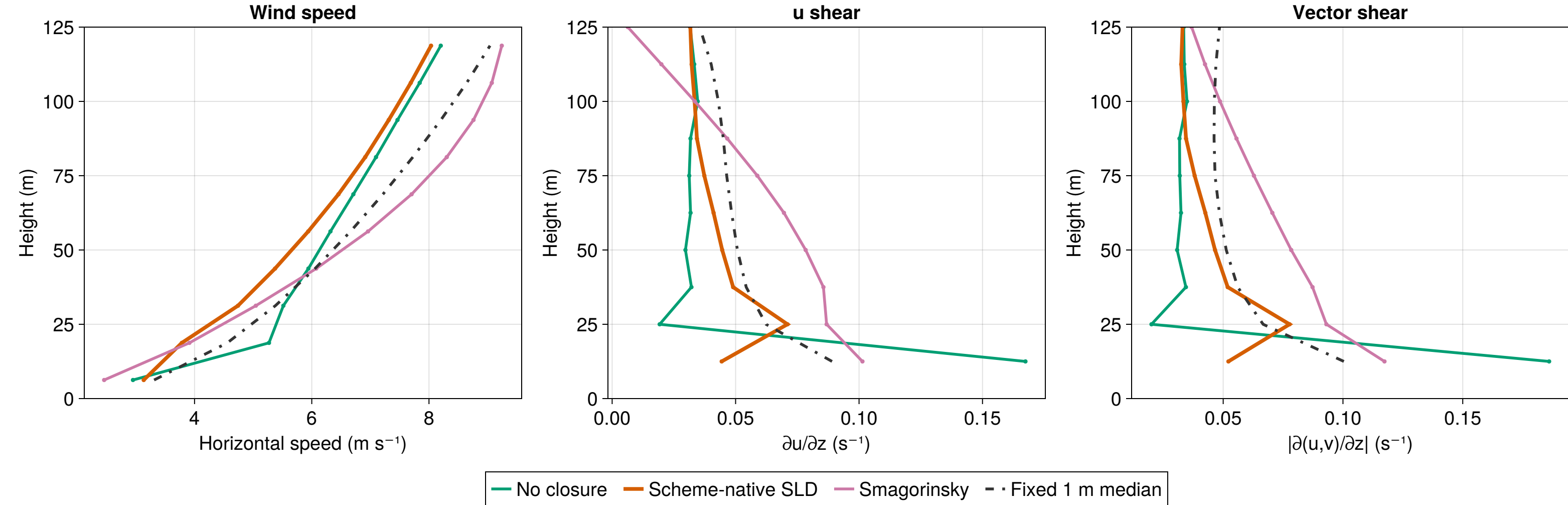
12.5 m WENO9 · same 400 m cube · derivatives between adjacent cell-center heights



Reference is for 8–9 h only: componentwise median u and v profiles sampled at 12.5 m cell centers, then differentiated identically.
The reference shear is not the median of member shears. Each curve is a layer slope at 12.5, 25, 37.5... m, not a wall gradient.

GABLS1 mean wind and shear: three requested treatments

8–9 h · 12.5 m WENO9 · 32^3 · same 400 m cube; fixed 1 m LES median for context

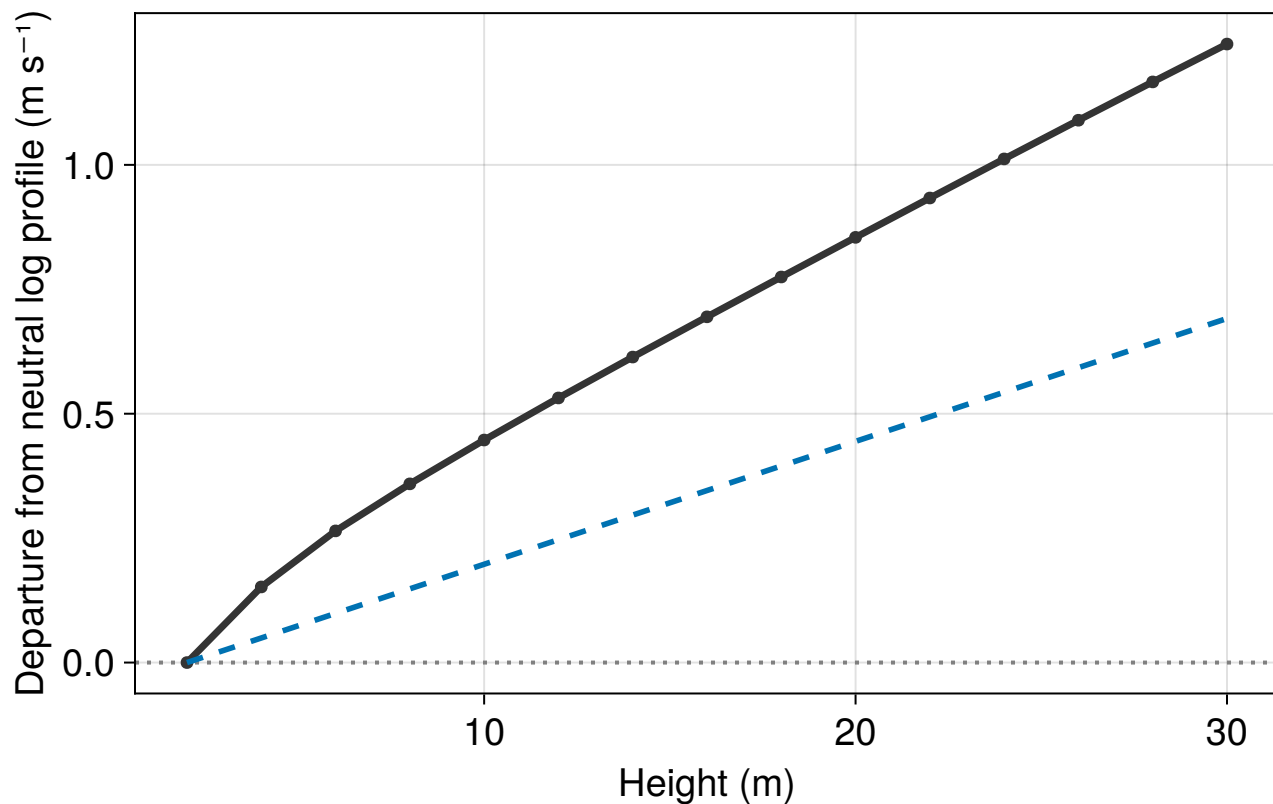


Reference componentwise median u/v sampled at 12.5 m centers before differentiation. Shear at 12.5 m spans 6.25–18.75 m; it is not a wall gradient.
Smagorinsky used an earlier source revision and is contextual rather than an exact same-revision treatment.

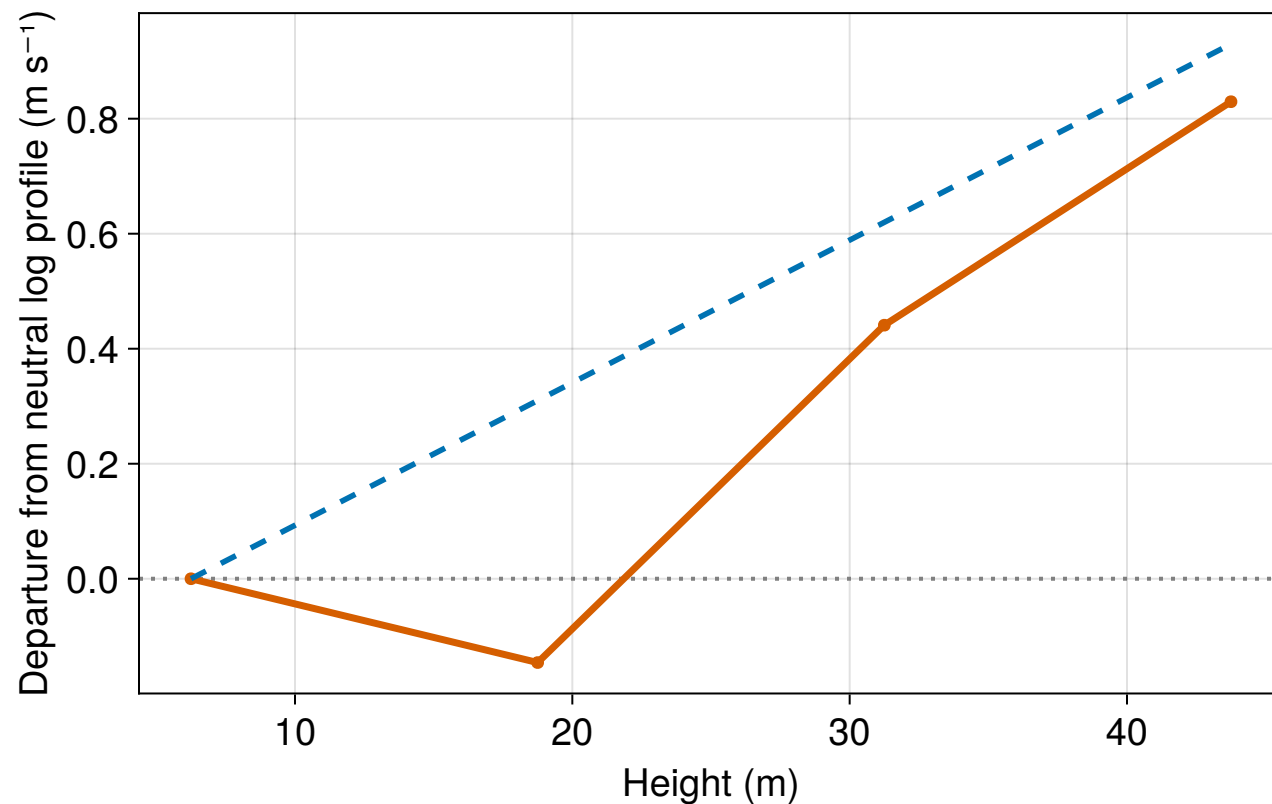
Neutral log-law versus stable surface-layer scaling

Residual from a neutral log profile anchored at the first plotted level; zero would be neutral-log behavior

Fixed 1 m LES median · 2–30 m



Scheme-native SLD · 6.25–43.75 m



···· Neutral log-law — Saved wind profile - - Stable MOST term

MOST term = $(u^*/\kappa) \beta_m(z-z_1)/L$, $\beta_m=4.8$; each curve uses its mean 8–9 h u^* and L . Reference L is derived from median u^* and heat flux.
Averaged profiles and fluxes need not satisfy pointwise MOST exactly; the law is a surface-layer expectation, not a constraint imposed through the column.